

Computational Simulation for Theory Development

SENSS Specialist Training: Introduction to Computational Methods

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Part 3

What is Computation as Theory?

- Abstract definitions are hard to grasp at first
- Instead, let's start with an applied example
- Schelling's Segregation Model
 - Agent-based model
 - Complexity
 - Emergence



https://en.wikipedia.org/wiki/The_Kongour_o_from_New_Holland

Racial Segregation in the US cities

- One of the first computational models ever developed
- Thomas Schelling in 1969
- 2005 Nobel Memorial Prize

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DYNAMIC MODELS OF SEGREGATION†

THOMAS C. SCHELLING
Harvard University

Some segregation results from the practices of organizations, some from specialized communication systems, some from correlation with a variable that is non-random; and some results from the interplay of individual choices. This is an abstract study of the interactive dynamics of discriminatory individual choices. One model is a simulation in which individual members of two recognizable groups distribute themselves in neighborhoods defined by reference to their own locations. A second model is a simulation in which individual members of two recognizable groups distribute themselves in neighborhoods defined by reference to their own locations. A second model is a simulation in which individual members of two recognizable groups distribute themselves in neighborhoods defined by reference to their own locations. A final section applies the analytics to the results. Exaggerated separation and patterning about individual motives can usually not be drawn from the results. Exaggerated separation and patterning about individual motives can usually not be drawn from the results. Exaggerated separation and patterning about individual motives can usually not be drawn from the results.



Racial Segregation in the US

Obvious reasons for segregation:

- institutional racism
- historical discrimination and injustices
- housing and zoning discrimination
- access to loans, etc.

Racial segregation is 'sticky':

- many policies to overcome it failed
- even when we tackle some of the institutional reasons, segregation remains

Can benign individual preferences contribute to racial segregation?



Can benign individual preferences contribute to racial segregation?

A thought exercise:

- Two identities: red and blue
- Individuals not overly racist:
 - they are happy to live next to the other identity
- They are even happy to be a minority in their neighbourhood
- They just don't want to be a too small of a minority
 - Only red in a blue neighbourhood
 - Less than a quarter ($25\% <$ is a problem, but $25\% \geq$ is fine.)
- If they are unhappy (too small of a minority), they can move

Can this lead to racial segregation?

What do you think?

1. Can such benign preferences lead to racial segregation?
2. What percentage threshold is sufficient for such preferences to start producing segregation? *(Enter a number between 0 and 100)*
3. What would your own threshold preference be? *(Enter a number between 0 and 100)*
4. What is the average threshold preference in the society in which we live? *(Enter a number between 0 and 100)*



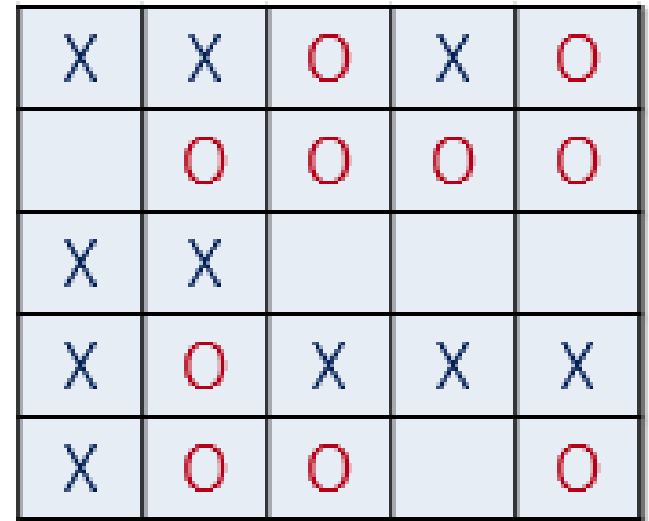
[Racial Segregation Questions – Fill in form](#)

How to tackle this question?

- Build an artificial world
- A simple computational model

Computational Model

- The city is represented by a grid: each grid-cell is a house (a location to live)
- Individual occupants (agents) live in a grid-cell
- Two types of racial identities: X (blue) and O (red)
- Empty cells represent vacant houses that agents can move in each round
- Initial condition: agents are placed randomly in the grid
- Grid-cells: red, blue or empty



A 5x5 grid representing a city layout. The grid contains blue 'X' and red 'O' characters, representing agents of different racial identities. Empty cells represent vacant houses. The grid is as follows:

X	X	O	X	O
	O	O	O	O
X	X			
X	O	X	X	X
X	O	O		O

Agent Satisfaction with Location

- An agent is satisfied with its location if **some** of its neighbours share its racial identity
 - **Some** is a *parameter* here: τ
- Let's say $\tau = 30\%$
 - an agent is fine with current location if at least 30% of its neighbours share its racial identity
 - a simplistic but plausible assumption
- **Individuals are not strictly racist**: they are happy living as a minority as long as it is not too tiny of a minority (less than 30 %)

Satisfied
because 1/2
(50%) of
neighbors
are X

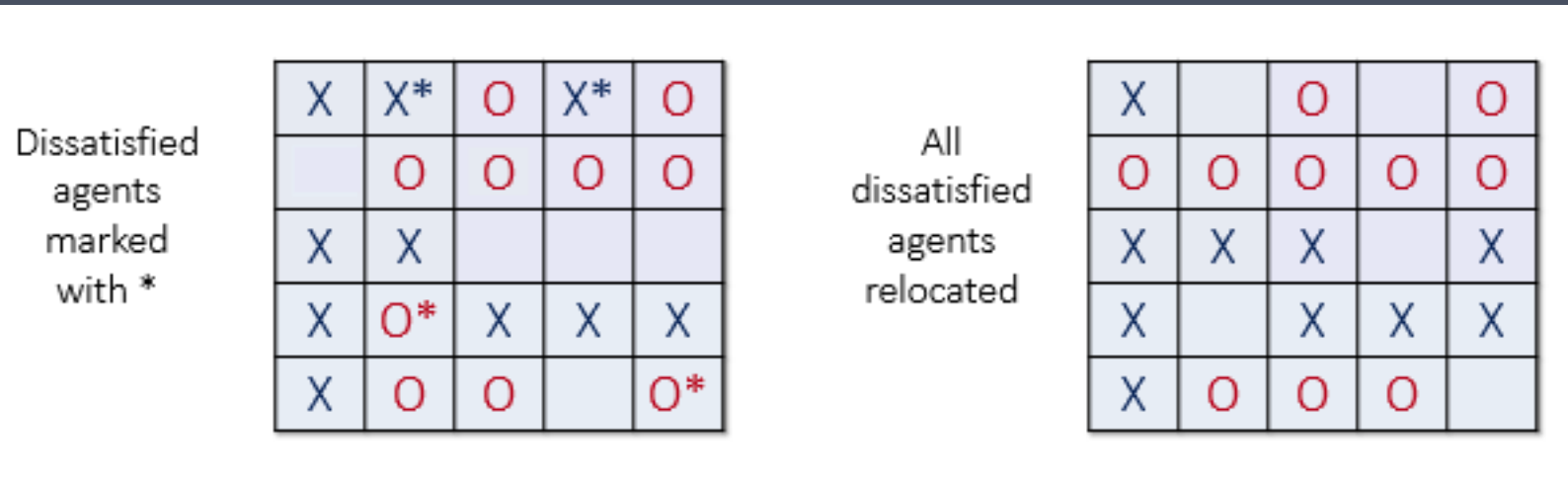
X	X	O	X	O
	O	O	O	O
X	X			
X	O	X	X	X
X	O	O		O

Dissatisfied because
only 1/4 (25%) of
neighbors are X

X	X	O	X	O
	O	O	O	O
X	X			
X	O	X	X	X
X	O	O		O

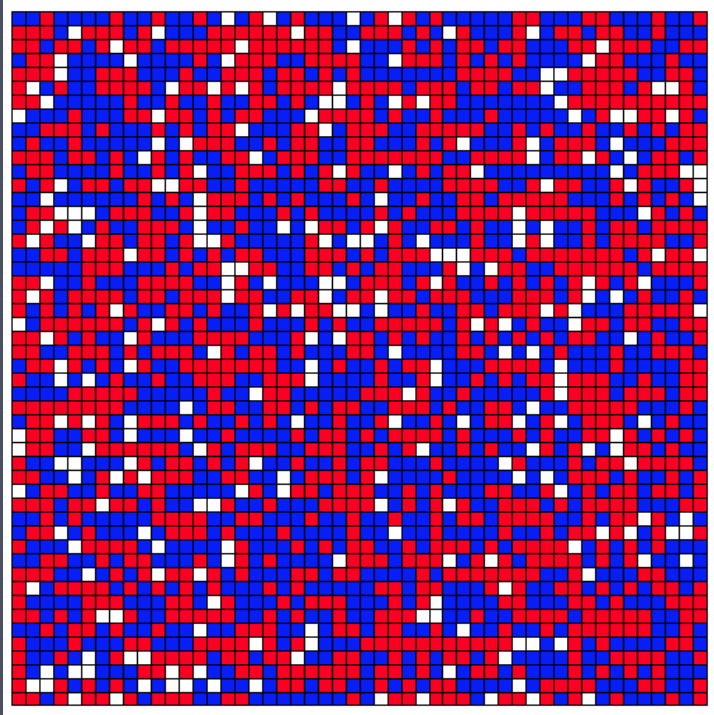
Agent Satisfaction with Location

- Dissatisfied agents are marked with an asterisks
- In each round, dissatisfied agents can move to a vacant cell
- Dissatisfied agents continue to move until they are satisfied

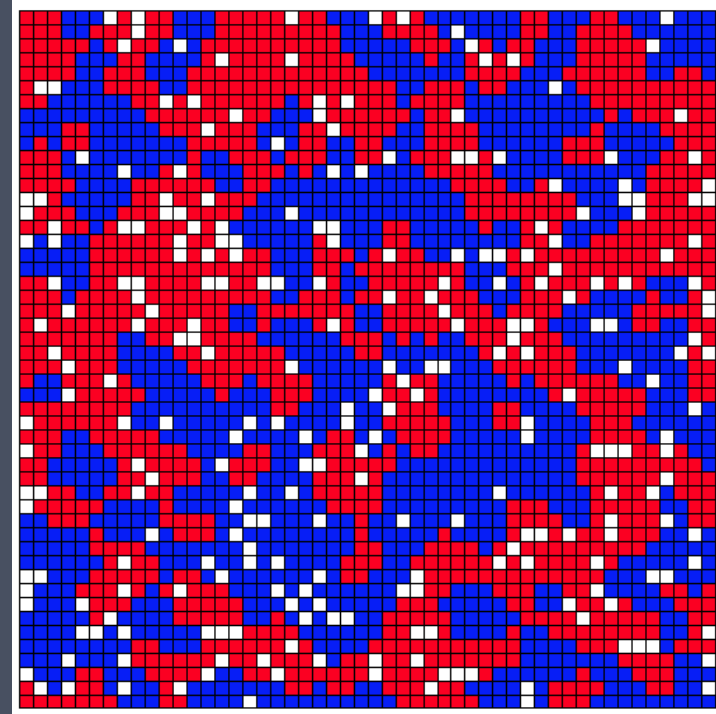


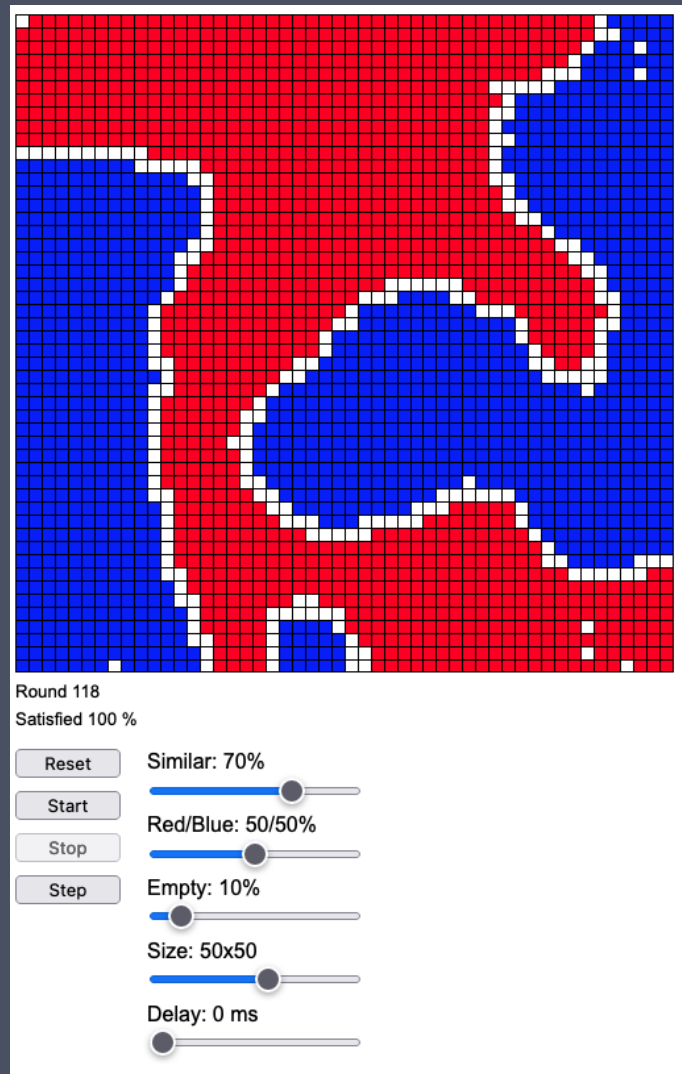
Racial Segregation in US Cities ($t = 30\%$)

Starting Condition



Outcome



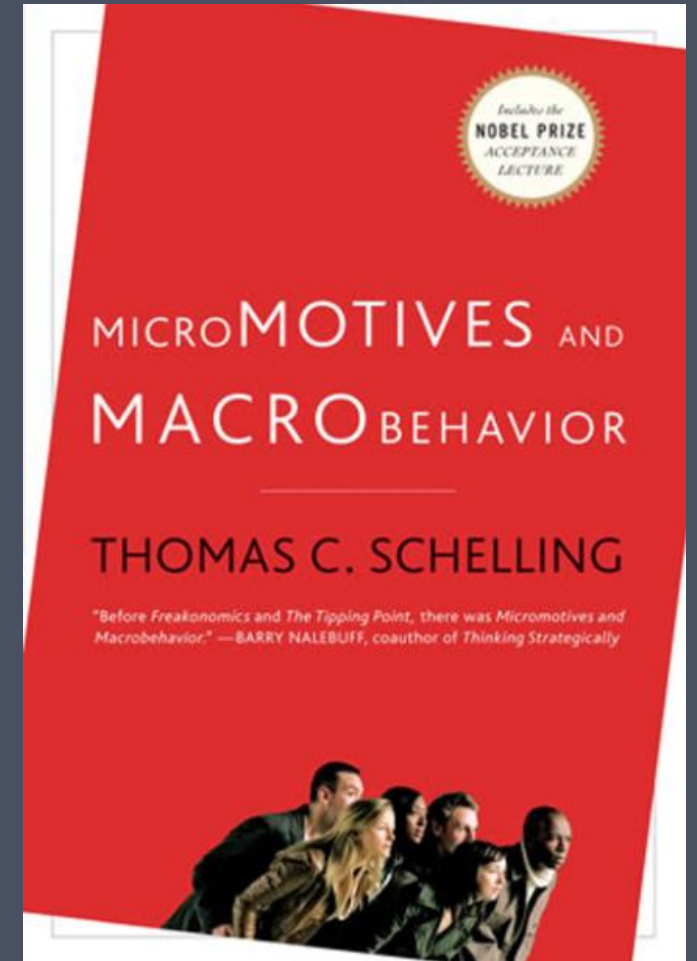


Explore the Model Yourself

- **Schelling's Model of Segregation explained by Frank McCown:**
http://nifty.stanford.edu/2014/mccown-schelling-model-segregation/?utm_source=pocket_mylist
- You can change the threshold
- You can see the different rounds

Emergence: Segregation

- **Micro-motives:** benign preference of not being too small of a minority
- Agents do not want or intend to create segregation
- Yet: repeated local decisions generate global segregation
- **Micro-motives** → macro-outcomes
- Segregation *emerge*



Emergence

- **Emergence** is a key concept in complex systems
- It is hard to define and grasp, but it is fascinating!
- Segregation appearing in Schelling's model is one example of *emergence*

Agent-Based Modelling

- Schelling's computational model: agent-based model
- A pioneering & defining example
- Simple but very powerful way to develop theory
- Which computer language / coding environment did Schelling use?

Agent-Based Modelling

- Computational models do not necessarily need electronic computers
- Schelling did not use computers: he used board and pieces

Agent-Based Modelling

- Create simple worlds with **agents** and surrounding **environment**
- **Agents** act following simple rules
- The goal: simplify and formalize actions and interactions of agents to understand how a system produces outcomes
- **Agents:** the units that act and interact with each other and their surrounding environment
 - individuals, firms (companies), states, armies/rebel groups, political actors, etc.

Explore Yourself

- See NetLogo! is a simple yet powerful environment to create agent
- Go to the Web Version and explore the Schelling's Segregation Model: <https://www.netlogoweb.org/>
- Also consider downloading NetLogo Desktop:
<https://www.netlogo.org/download/>
 - It is free!
 - Extremely useful for learning!